

Supplementary Material

I. PIECEWISE LINEAR REFORMULATION OF THE QUADRATIC ANTENNA GAIN MODEL

To preserve the mixed-integer linear programming (MILP) structure of the proposed optimization model, the quadratic antenna gain term $\vartheta_{isut}^2 = (\vartheta_{isut}^1)^2$ is reformulated using a piecewise linear outer approximation based on secant functions. The construction follows the optimal secant approximation proposed in [1]. For completeness, the corresponding result is reproduced below.

Theorem S1 ([1]). *Let $x \in [x_{\min}, x_{\max}]$ with $x_{\min} > 0$ be a nonnegative continuous variable and η the number of secant segments. The secants that achieve the tightest convex outer approximation of $y = x^2$ with maximum relative error at most ϵ are given by*

$$k_p = (1 + 2\epsilon + 2\sqrt{\epsilon + \epsilon^2})^{p-1} (1 + 2\epsilon + 2\sqrt{\epsilon + \epsilon^2}) x_{\min},$$

$$b_p = - \left(1 + 2\epsilon + 2\sqrt{\epsilon + \epsilon^2}\right)^{2p-1} x_{\min}^2, \quad p = 1, 2, \dots, \eta,$$

where the minimum number of required segments is

$$\eta = \left\lceil \frac{\ln(x_{\max}/x_{\min})}{\ln \kappa} \right\rceil, \quad \kappa = 1 + 2\epsilon + 2\sqrt{\epsilon + \epsilon^2}.$$

II. ADAPTIVE PIECEWISE LINEARIZATION PROCEDURE

This supplementary section provides the detailed procedure used to generate the piecewise linear approximation parameters adopted in the main manuscript. The logarithmic power-conversion function is given by

$$f(x) = 10 \log_{10}(x), \quad (S1)$$

where x denotes the transmit power in the linear scale. Since this function is nonlinear, it is approximated by a set of linear segments over the predefined power domain $[x_s, x_e]$. For each segment p , the approximation is written as

$$f(x) \approx k_p x + b_p, \quad x \in [x_p, x_{p+1}], \quad (S2)$$

where x_p and x_{p+1} are the lower and upper endpoints of segment p , respectively, and k_p and b_p are the corresponding slope and intercept.

The segment parameters are obtained using a secant-based approximation. Specifically, for a given interval $[x_p, x_{p+1}]$, the slope and intercept are calculated as

$$k_p = \frac{f(x_{p+1}) - f(x_p)}{x_{p+1} - x_p}, \quad (S3)$$

$$b_p = f(x_p) - k_p x_p. \quad (S4)$$

To control the approximation accuracy, the maximum relative error of each segment is evaluated as

$$e_p^{\max} = \max_{x \in [x_p, x_{p+1}]} \frac{|k_p x + b_p - f(x)|}{|f(x)|}. \quad (S5)$$

In practice, this maximum error is computed over a finite set of sampled points within the interval. If $e_p^{\max}$ exceeds the prescribed tolerance ϵ , the segment length is reduced and the corresponding slope and intercept are recomputed. Conversely, if the approximation error is sufficiently small, the segment length is increased for the next interval to reduce the total number of segments.

The generated segment endpoints (x_p, x_{p+1}) , slopes k_p , and intercepts b_p are then used in the main optimization model. In particular, the binary segment-selection variable activates exactly one segment for each power variable, and the big- $\mathcal{M}$ constraints enforce the corresponding linear relationship between the linear-scale transmit power and its dB-scale approximation.

Algorithm S1 Adaptive Piecewise Linearization

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1: Input: function to linearize,  $f$ ; starting point of domain,  $x_s$ ; ending point
   of domain,  $x_e$ ; initial step size,  $\Delta_0 = 0.5$ ; step decrease factor,  $\beta_d = 0.5$ ;
   step increase factor,  $\beta_i = 1.5$ ; maximum allowed relative error,  $\epsilon = 1\%$ .
2: Output: List of linear segments  $(x_p, x_{p+1}, k_p, b_p, e_p^{\max})$  for  $p = 1, \dots, P$ 
3: procedure ADAPTIVELINEARIZATION( $f, x_s, x_e, \epsilon$ )
4:    $x_c \leftarrow x_s$ , segments  $\leftarrow []$ ,  $\Delta \leftarrow \Delta_0$ ,  $p \leftarrow 1$ 
5:   while  $x_c < x_e$  do
6:      $x_p \leftarrow x_c$  ▷ Store current starting point
7:      $x_{p+1} \leftarrow \min(x_c + \Delta, x_e)$ 
8:      $(k_p, b_p) \leftarrow \text{Secant}(x_p, x_{p+1}, f)$ 
9:      $e_p^{\max} \leftarrow \text{MaxError}(x_p, x_{p+1}, f, k_p, b_p)$ 
10:    while  $e_p^{\max} > \epsilon$  do ▷ Refine if error too large
11:       $\Delta \leftarrow \Delta \times \beta_d$  ▷ Decrease step
12:       $x_{p+1} \leftarrow x_p + \Delta$ 
13:       $(k_p, b_p) \leftarrow \text{Secant}(x_p, x_{p+1}, f)$ 
14:       $e_p^{\max} \leftarrow \text{MaxError}(x_p, x_{p+1}, f, k_p, b_p)$ 
15:    end while
16:    segments.add( $x_p, x_{p+1}, k_p, b_p, e_p^{\max}$ )
17:    if  $e_p^{\max} < 0.5\epsilon$  then ▷ Expand if highly accurate
18:       $\Delta \leftarrow \Delta \times \beta_i$ 
19:    end if
20:     $x_c \leftarrow x_{p+1}$  ▷ Move to next segment
21:     $p \leftarrow p + 1$ 
22:  end while
23:  return segments
24: end procedure
25: procedure SECANT( $x_p, x_{p+1}, f$ )
26:    $y_p \leftarrow f(x_p)$ ,  $y_{p+1} \leftarrow f(x_{p+1})$ 
27:    $k_p \leftarrow (y_{p+1} - y_p) / (x_{p+1} - x_p)$  ▷ Slope of segment  $p$ 
28:    $b_p \leftarrow y_p - k_p x_p$  ▷ Intercept of segment  $p$ 
29:   return  $(k_p, b_p)$ 
30: end procedure
31: procedure MAXERROR( $x_p, x_{p+1}, f, k_p, b_p$ )
32:    $e_p^{\max} \leftarrow 0$ 
33:   for  $x \in \text{Sample}(x_p, x_{p+1}, N)$  do
34:      $y_{\text{true}} \leftarrow f(x)$ ,  $y_{\text{approx}} \leftarrow k_p x + b_p$ 
35:      $e \leftarrow |y_{\text{approx}} - y_{\text{true}}| / |y_{\text{true}}|$ 
36:      $e_p^{\max} \leftarrow \max(e_p^{\max}, e)$ 
37:   end for
38:   return  $e_p^{\max}$ 
39: end procedure

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The complete adaptive piecewise linearization procedure is summarized in Algorithm S1. After the adaptive segmentation process is completed, the resulting set of linear segments

is applied consistently to the corresponding transmit-power variables in the main manuscript. For the MaBS transmit-power conversion, the segment parameters are denoted as $(x_p^2, x_{p+1}^2, k_p^2, b_p^2)$, $p \in \mathcal{P}_2$, and are incorporated into the piecewise linearization constraints presented in the main manuscript. The same procedure is applied to the MiBS transmit-power conversion.

III. COMPUTATIONAL COMPLEXITY

We analyze the computational complexity of the proposed ALNS-PR algorithm for a single time slot. Let N denote the number of UEs, S the number of BSs, U the number of sectors per macro BS, K the maximum number of cached candidate BSs per UE, α the removal ratio, and I the maximum number of ALNS iterations. In each iteration, approximately αN UEs are removed and then reassigned.

The algorithm consists of three main parts: preprocessing, ALNS iterative search, and final power/down-tilt refinement.

Preprocessing. Before the iterative search starts, the algorithm constructs the UE-BS feasibility matrix and the candidate-BS cache. The feasibility matrix checks whether each UE can be served by each BS and, for macro BSs, by each sector. This step requires $\mathcal{O}(NSU)$ time. In addition, for each UE, feasible candidate BSs are sorted and only the top K candidates are stored. Sorting the candidates costs $\mathcal{O}(S \log S)$ per UE, and thus candidate caching requires $\mathcal{O}(NS \log S)$ time. Therefore, the overall preprocessing complexity is $\mathcal{O}(NSU + NS \log S)$.

Destroy phase. In each ALNS iteration, one destroy operator is selected adaptively. Although different destroy operators follow different removal rules, they mainly scan the current UE-BS assignment and update the states of the affected BSs/sectors. Scanning the current assignment costs $\mathcal{O}(NSU)$ in the worst case. Moreover, approximately αN UEs are removed, and updating the affected BS/sector states may require checking the remaining served UEs, which costs $\mathcal{O}(N)$ per removed UE. Hence, the destroy phase is bounded by $\mathcal{O}(NSU + \alpha N^2)$.

Repair phase. The repair phase is the dominant part of each iteration. After the destroy phase, approximately αN removed UEs must be reassigned. For each removed UE, the algorithm examines at most K cached candidate BSs. For each candidate BS, feasibility and bandwidth-capacity checks are performed. In the worst case, these checks require scanning the current UE assignment, resulting in $\mathcal{O}(N)$ time per candidate. Consequently, the repair phase requires $\mathcal{O}(\alpha N \cdot K \cdot N) = \mathcal{O}(\alpha K N^2)$ time per iteration.

Other operations in one iteration, such as solution copying, feasibility checking, objective evaluation, and acceptance decision, have lower asymptotic orders compared with the destroy and repair operations. Therefore, the complexity of one ALNS iteration is bounded by $\mathcal{O}(NSU + \alpha N^2 + \alpha K N^2)$.

Final refinement. After the ALNS iterative search terminates, the best solution is further refined by optimizing the transmit power and antenna down-tilt angles. This step scans the active BSs and their served UEs. For macro BSs, the sector-level configurations are also considered. Therefore, the final refinement step costs $\mathcal{O}(SUN)$ time.

Combining preprocessing, the iterative search, and the final refinement, the total time complexity for a single time slot is $\mathcal{O}(NSU + NS \log S + I(NSU + \alpha N^2 + \alpha K N^2) + SUN)$. (S6)

Since U , K , and α are fixed constants in the implementation, the above expression can be simplified as

$$\mathcal{O}(NS \log S + I(NS + N^2) + SN). \quad (S7)$$

In practical large-scale scenarios, the iterative search usually dominates the one-time preprocessing and final refinement steps. Therefore, the leading practical time complexity is $\mathcal{O}(IN^2)$.

IV. HYPERPARAMETER OPTIMIZATION AND SENSITIVITY ANALYSIS

The hyperparameters of ALNS-PR were calibrated using Optuna with the Tree-structured Parzen Estimator (TPE) sampler [2], [3] on the small-scale scenario described in Section VI-B.1. TPE is a sequential model-based optimization algorithm that iteratively updates a probabilistic model of the hyperparameter search space based on previously evaluated configurations, thereby guiding the search toward promising regions. The optimization objective was to minimize the final network power consumption obtained by ALNS-PR.

The search space covered the maximum number of iterations, segment length, stagnation threshold, removal rate, reaction factor, simulated annealing parameters, and adaptive reward coefficients. The best configuration obtained from the small-scale TPE calibration was reported in Table IV of the main manuscript and was then fixed for all subsequent experiments in Sections VI-B.2)–VI-B.5). No additional per-instance or per-scenario tuning was performed in the large-scale experiments. This design ensures that the reported performance is evaluated under a single transferable parameter configuration, rather than under case-specific hyperparameter optimization.

To assess whether the TPE-calibrated configuration in Table IV of the main manuscript remains robust after being transferred to larger experimental settings, we further conducted a one-factor-at-a-time (OFAT) sensitivity analysis. In each OFAT experiment, only one hyperparameter was varied within its predefined range, while all remaining parameters were fixed. The final network power consumption was adopted as the primary evaluation metric, and the corresponding energy-saving ratio with respect to the All-On baseline was also recorded.

Table S1 summarizes the sensitivity results. For each hyperparameter, the reported power range corresponds to the minimum and maximum average power consumption over all tested values, and the maximum deviation is computed relative to the TPE-calibrated configuration in Table IV of the main manuscript. Overall, ALNS-PR exhibits low sensitivity to moderate hyperparameter perturbations. The largest deviation from the calibrated configuration is only 1.875%, while the majority of hyperparameters produce deviations below 0.7%. This demonstrates that the selected parameter configuration is stable and does not rely on extensive fine-grained tuning.

TABLE S1: Summary of OFAT hyperparameter sensitivity results.

Parameter	Default	Tested values	Power range (kW)	Max deviation (%)
$N_{\max}$	200	100, 200, 300	24.336–24.336	0.00
N_{segment}	20	10, 20, 40	24.336–24.379	0.18
$N_{\text{stagnation}}$	50	25, 50, 100	24.336–24.498	0.67
α_{remove}	0.20	0.10, 0.20, 0.30	24.232–24.792	1.88
λ	0.30	0.10, 0.30, 0.50	24.326–24.381	0.19
T_0	1000	500, 1000, 1500	24.280–24.489	0.63
α_{cool}	0.995	0.990, 0.995, 0.999	24.299–24.489	0.63
$T_{\min}$	1.0	0.5, 1.0, 2.0	24.336–24.336	0.00
ω_{best}	40	30, 40, 60	24.326–24.477	0.58
ω_{improve}	15	10, 15, 25	24.336–24.483	0.61
ω_{accept}	5	3, 5, 10	24.336–24.498	0.67
ω_{reject}	-8	-12, -8, -5	24.289–24.489	0.63

Among all hyperparameters, the removal rate α_{remove} exhibits the highest sensitivity. A smaller value limits exploration, whereas a larger value slightly improves solution quality at the cost of additional computation. Consequently, the default value $\alpha_{\text{remove}} = 0.20$ provides a good trade-off between solution quality and computational efficiency. The remaining hyperparameters exhibit only marginal influence on the final solutions. In particular, the reaction factor, simulated annealing parameters, and adaptive reward coefficients all lead to relatively small performance variations, indicating that the adaptive operator selection and acceptance mechanisms remain effective over a broad range of parameter settings.

These results confirm that the hyperparameters calibrated on the small-scale scenario transfer well to the larger experimental settings. Therefore, the reported performance of ALNS-PR is obtained under a single fixed parameter configuration and does not rely on fragile or scenario-specific fine tuning.

REFERENCES

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